Lab 3: IMU Noise Characterization with Allan Variance

EECE 5554 | Nishka Shastri

1. Driver and Data Conversions

A ROS 2 driver (`imu_driver/driver.py`) was written for the VN-100. It opens the serial port at 115200 baud and sets the output rate by writing to the IMU registers instead of sleeping in the code: `$VNWRG,06,14` sets the output type to VNYMR and `$VNWRG,07,40` sets the rate to 40 Hz. `ros2 topic hz /imu` showed about 40.0 Hz. Just sleeping in the driver would not work, because the IMU would still send data at its own rate and the extra data would pile up in the serial buffer. Each $VNYMR line is checked with its checksum (XOR of all characters between $ and *), and lines with a wrong checksum are skipped. Valid lines are split into yaw, pitch, roll, magnetometer, accelerometer and gyro values. The data is published on topic `imu` as a custom `IMUmsg` (header with frame_id `IMU1_Frame` and sec/nanosec time stamp, `sensor_msgs/Imu`, `sensor_msgs/MagneticField`, and the raw string). The port is given at run time through the launch file, for example `ros2 launch imu_driver imu_launch.py port:=/dev/ttyUSB0`.

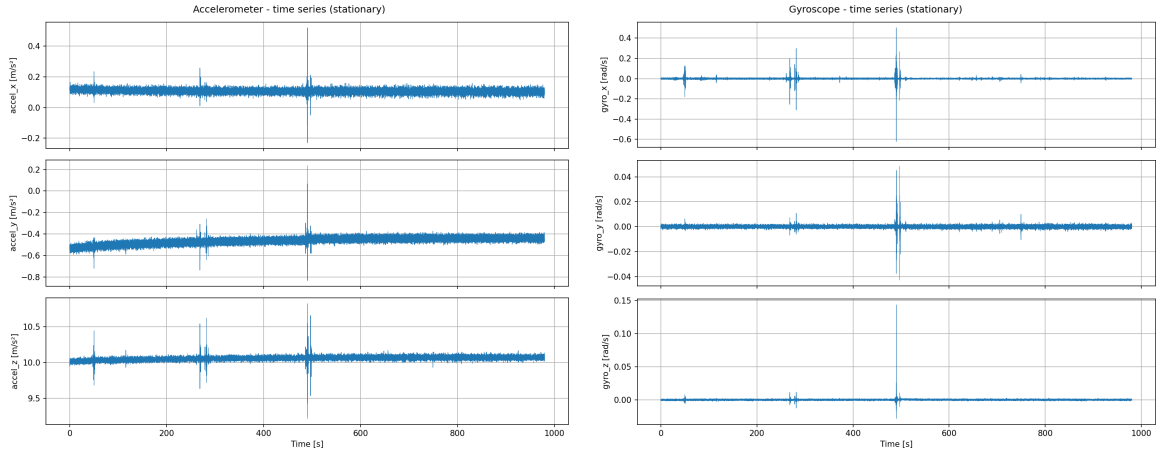
Conversions: angles were changed from degrees to radians and converted to a quaternion (ZYX order, roll φ, pitch θ, yaw ψ) without libraries:

$$w = c\varphi\, c\theta\, c\psi + s\varphi\, s\theta\, s\psi,\quad x = s\varphi\, c\theta\, c\psi - c\varphi\, s\theta\, s\psi,\quad y = c\varphi\, s\theta\, c\psi + s\varphi\, c\theta\, s\psi,\quad z = c\varphi\, c\theta\, s\psi - s\varphi\, s\theta\, c\psi \quad (c = \cos(\cdot/2),\ s = \sin(\cdot/2))$$

The magnetometer was converted from Gauss to Tesla ($\times 10^{-4}$). Acceleration (m/s²) and angular rate (rad/s) were already in ROS units. For the analysis, the quaternions were converted back to Euler angles with $\varphi = \text{atan2}(2(wx+yz), 1-2(x^2+y^2))$, $\theta = \text{asin}(2(wy-zx))$, $\psi = \text{atan2}(2(wz+xy), 1-2(y^2+z^2))$.

2. Stationary Data (Individual)

About 18 minutes of data was recorded with the IMU stationary (mounted upside down, so roll ≈ 177° and accel Z ≈ +10 m/s²). Motion was found by checking where the gyro magnitude was above 0.01 rad/s. The start and end, where the IMU was being placed and picked up, were cut, leaving 980 s (39,201 samples). The bag was converted to CSV and analyzed in Python. Figure 1 shows the time series.



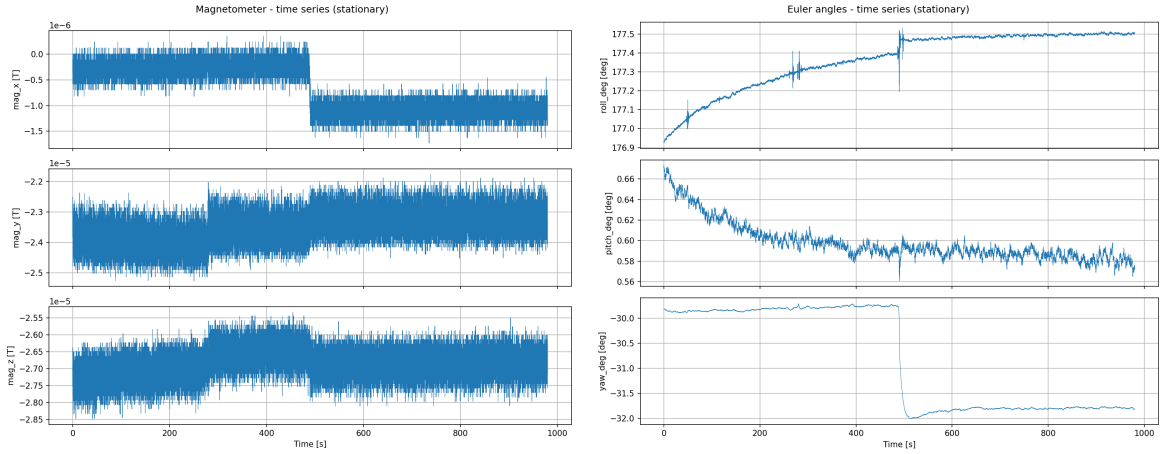


Figure 1: Time series of accelerometer, gyroscope, magnetometer and Euler angles (stationary).

The data has a few disturbances (bumps) at about 50 s, 270–285 s and 490 s, mostly about the X axis. Roll and pitch also drift slowly during the first 490 s, and accel Y changes by about 0.1 m/s² over the same time, which matches the roll change (g·sin(0.57°) ≈ 0.1 m/s²), so the IMU was actually tilting slightly. The bump at 490 s rotated the IMU, which caused a 2° step in yaw and steps in the magnetometer.

Table 1: Mean and standard deviation of the stationary data.

Signal	Mean	Std. dev.	Signal	Mean	Std. dev.
Accel X (m/s²)	0.1066	0.0153	Mag X (µT)	−0.677	0.447
Accel Y (m/s²)	−0.4619	0.0341	Mag Y (µT)	−23.49	0.545
Accel Z (m/s²)	10.0598	0.0300	Mag Z (µT)	−26.86	0.449
Gyro X (rad/s)	−3.40e−5	8.73e−3	Roll (deg)	177.366	0.150
Gyro Y (rad/s)	7.63e−5	1.10e−3	Pitch (deg)	0.598	0.019
Gyro Z (rad/s)	1.02e−5	1.28e−3	Yaw (deg)	−30.807	1.006

The gyro means are close to zero (small bias). The accelerometer magnitude is 10.07 m/s², slightly more than g, because of accelerometer bias and scale errors (no calibration was done). The gyro means are of the same order as the Earth's rotation rate (7.29e−5 rad/s), so the bias cannot be separated from it with this short recording. Gyro X has a much larger standard deviation than Y and Z because of the bumps. The magnetic field magnitude is about 35.7 µT, lower than the expected Earth field (around 50 µT), because the magnetometer was not calibrated and indoor fields are distorted by nearby metal and electronics.

The standard deviations can also be compared with the white noise level found in the Allan variance section. For white noise with density N sampled at 40 Hz, the expected standard deviation is about N·√(20 Hz). This gives about 4.6e−4 rad/s for the gyro and 8.5e−3 m/s² for accel X. The measured values are about 2–3 times larger, which again shows that the bumps and the slow tilt added extra variation on top of the sensor noise. A quieter location (for example a basement floor) would give values closer to the pure sensor noise.

Figure 2 shows the histograms of each signal. To make the shape visible, the x-axis covers the 0.5 to 99.5 percentile range, so the few large bump values do not stretch the plot. The red curve is a Gaussian with the same mean and standard deviation as the data.

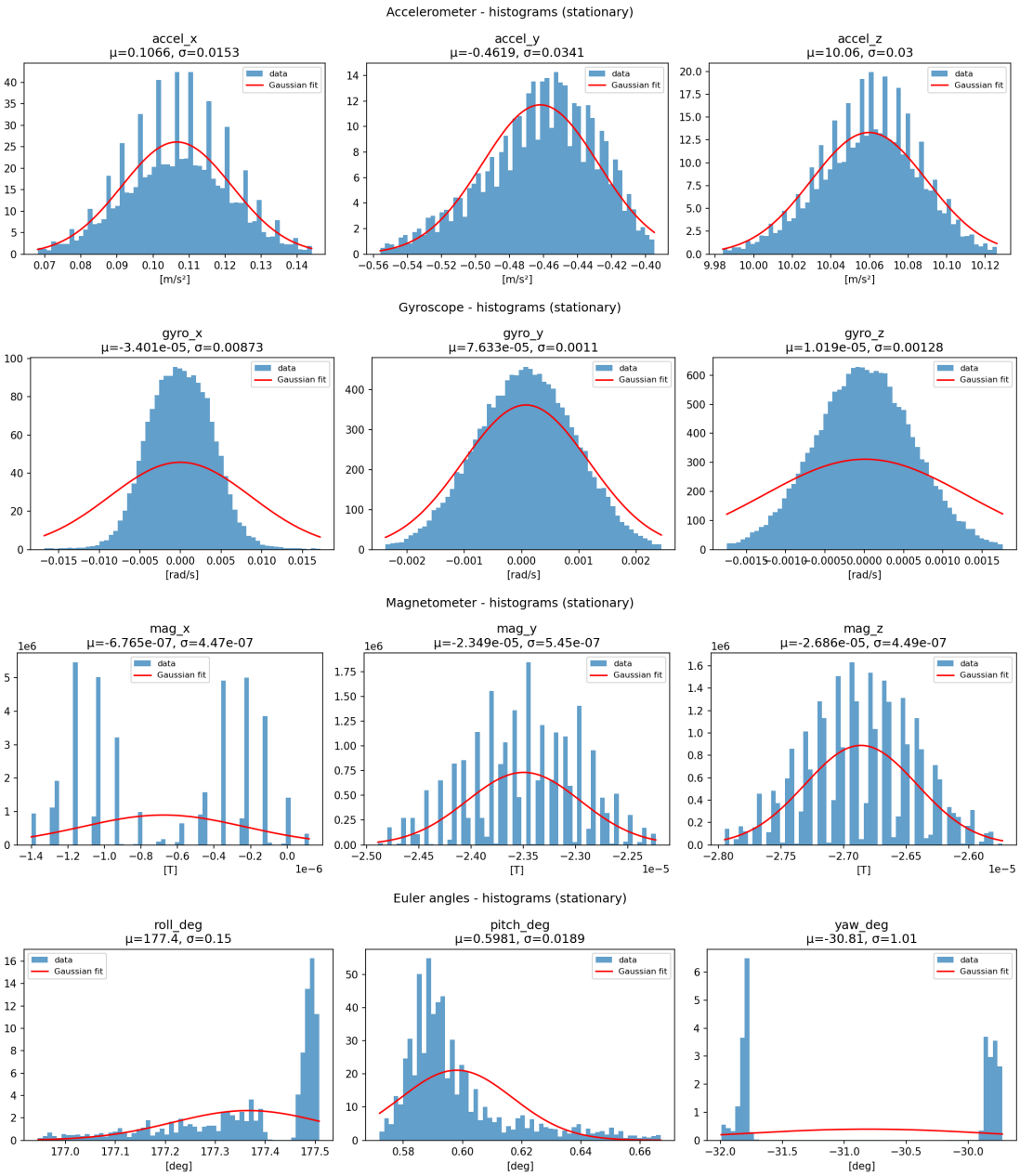


Figure 2: Histograms (0.5–99.5 percentile range) with a Gaussian fit using the sample mean and std. dev.

Accelerometer. All three axes follow a Gaussian distribution well. This is expected, since sensor noise comes from many small independent sources (central limit theorem). The comb pattern in the bars is because the output only has 0.001 m/s² resolution, which is close to the bin width. Accel Y is slightly skewed because of the slow tilt.

Gyroscope. The gyro is also bell-shaped around zero. For X and Z, the data is more peaked than the Gaussian fit, because the few large bump values increase the standard deviation and make the fitted curve too wide. Gyro Y, which had the fewest disturbances, fits the Gaussian well. So the actual sensor noise is Gaussian, and the outliers come from the environment.

Magnetometer. The magnetometer histograms have separate spikes because its output is quantized (it can only take certain values). Y and Z still roughly follow a Gaussian shape. X has two groups because its value shifted at 490 s.

Euler angles. The orientation angles are not Gaussian. Roll has a long tail because of the slow drift, pitch is skewed, and yaw has two peaks about 2° apart from the step at 490 s. These shapes come from the IMU slightly moving, not from random noise. In a quiet 56 s part of the data, roll and pitch had standard deviations of only 0.004° and 0.006°.

3. Allan Variance (Team Data)

About 5 hours of stationary data at 40 Hz was analyzed in MATLAB using the MathWorks Allan variance example. The rate data is integrated, split into clusters of length τ, and the Allan variance is computed from the differences between neighboring cluster averages for many values of τ. The Allan deviation is plotted on a log-log scale, where each noise type has its own slope. Figure 3 shows the Allan deviation for X, Y and Z, and Table 2 shows the results.

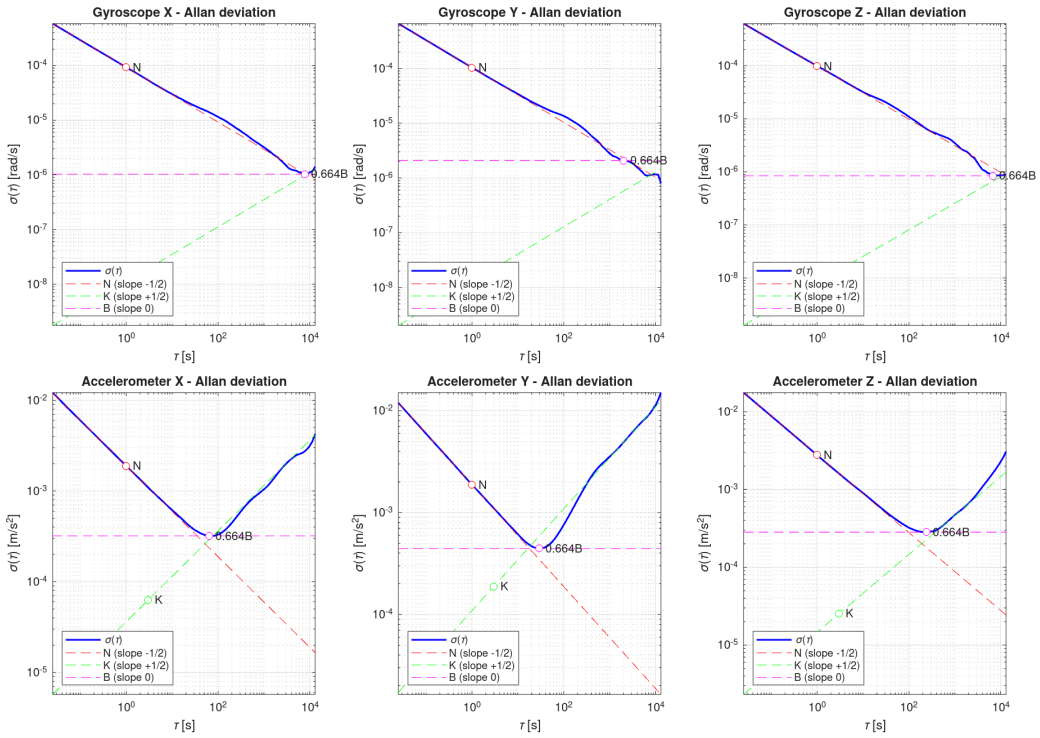


Figure 3: Allan deviation of the gyroscope (top) and accelerometer (bottom) for X, Y and Z.

Table 2: Noise parameters from the Allan deviation.

Axis	N	B	K
Gyro X	9.22e−5 rad/s/√Hz (0.317 °/√hr)	1.55e−6 rad/s (0.32 °/hr)	not observed
Gyro Y	1.04e−4 rad/s/√Hz (0.356 °/√hr)	3.13e−6 rad/s (0.65 °/hr)	not observed
Gyro Z	9.79e−5 rad/s/√Hz (0.337 °/√hr)	1.27e−6 rad/s (0.26 °/hr)	not observed
Accel X	1.89e−3 m/s²/√Hz (193 µg/√Hz)	4.81e−4 m/s² (49 µg)	6.28e−5 m/s³/√Hz
Accel Y	1.88e−3 m/s²/√Hz (191 µg/√Hz)	6.69e−4 m/s² (68 µg)	1.89e−4 m/s³/√Hz
Accel Z	2.78e−3 m/s²/√Hz (284 µg/√Hz)	4.26e−4 m/s² (43 µg)	2.54e−5 m/s³/√Hz

What kind of errors/sources of noise are present? The gyro is mostly white noise (angle random walk), since its curve keeps a −½ slope up to about 1000 s, and it only flattens (bias instability) near the end of the data. Rate random walk is not visible for the gyro. The accelerometer shows velocity random walk (−½ slope), bias instability (flat minimum at about 30–250 s) and rate random walk (+½ slope). Accel Y also rises faster at long times, which

looks like a rate ramp (slow drift, probably from temperature). Quantization noise (−1 slope) is not visible.

How do we model them, and where do we measure them? Angle/velocity random walk is white noise, modeled as $\sigma(\tau) = N/\sqrt{\tau}$, and N is read from the −½ line at $\tau = 1$ s. Bias instability is flicker noise in the bias; the curve is flat at $\sigma \approx 0.664B$, so B is the minimum divided by 0.664. Rate random walk is a random walk of the bias, $\sigma(\tau) = K\sqrt{(\tau/3)}$, and K is read from the +½ line at $\tau = 3$ s. A rate ramp gives a +1 slope.

Table 3: Noise types and how they appear on the Allan deviation plot.

Noise type	Cause / model	Slope	Where read
Angle / velocity random walk (N)	white noise on the rate output	−½	line value at $\tau = 1$ s
Bias instability (B)	flicker (1/f) noise in the bias	0	curve minimum / 0.664
Rate random walk (K)	bias changes as a random walk	+½	line value at $\tau = 3$ s
Rate ramp (R)	slow drift, e.g. temperature	+1	long τ region
Quantization (Q)	rounding of the digital output	−1	short τ (not seen here)

Comparison with the VN-100 datasheet. The datasheet gives a gyro noise density of 0.0035 °/s/√Hz (0.21 °/√hr), gyro in-run bias stability below 10 °/hr, accel noise density of 0.14 mg/√Hz (140 µg/√Hz) and accel bias stability below 0.04 mg (40 µg). The measured random walk is about 1.5–1.7 times the datasheet for the gyro and 1.4–2 times for the accelerometer, which is reasonable since the datasheet values are from ideal conditions and our data includes vibration and temperature changes. Accel bias instability (43–68 µg) is close to the datasheet. The gyro bias instability (0.26–0.65 °/hr) is better than the datasheet, probably because the VN-100 filter removes gyro bias from its output, and because there are very few clusters at τ of a few thousand seconds, so this value is not very reliable.

Table 4: Measured values compared with the VN-100 datasheet.

Parameter	Measured (X / Y / Z)	VN-100 datasheet	Ratio
Gyro angle random walk	0.317 / 0.356 / 0.337 °/√hr	0.21 °/√hr	1.5–1.7×
Gyro bias stability	0.32 / 0.65 / 0.26 °/hr	< 10 °/hr	below spec
Accel noise density	193 / 191 / 284 µg/√Hz	140 µg/√Hz	1.4–2.0×
Accel bias stability	49 / 68 / 43 µg	< 40 µg	1.1–1.7×

The accelerometer Z axis has the highest noise density. This axis is aligned with gravity and also picks up vertical vibrations from the floor. The small hump above the −½ line in the gyro plots between about 10 s and 1000 s may come from slow changes in the environment or from the internal filtering in the VN-100.

4. Conclusion

The driver publishes VN-100 data at 40 Hz with the correct units and quaternion orientation. The stationary accelerometer and gyro noise is roughly Gaussian, but the data was affected by bumps and slow tilting. The Allan variance shows that white noise is the main gyro error, and the accelerometer also has bias instability and rate random walk. The results are in the same range as the VN-100 datasheet.

This lab also showed how much the environment affects IMU data. Small bumps and slow tilting were clearly visible in the stationary data and increased the standard deviations, while the long team recording gave noise values close to the datasheet. For future recordings, the IMU should be fixed to a heavy surface in a quiet location, and the gyro bias instability would need a longer recording to be measured reliably. These noise parameters (N, B and K) can be used to set the noise values in a Kalman filter that combines the IMU with other sensors such as GPS.